Report

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23 February 2026

Notes

1 Initial Tests

2 Varying Number of Segments and Points

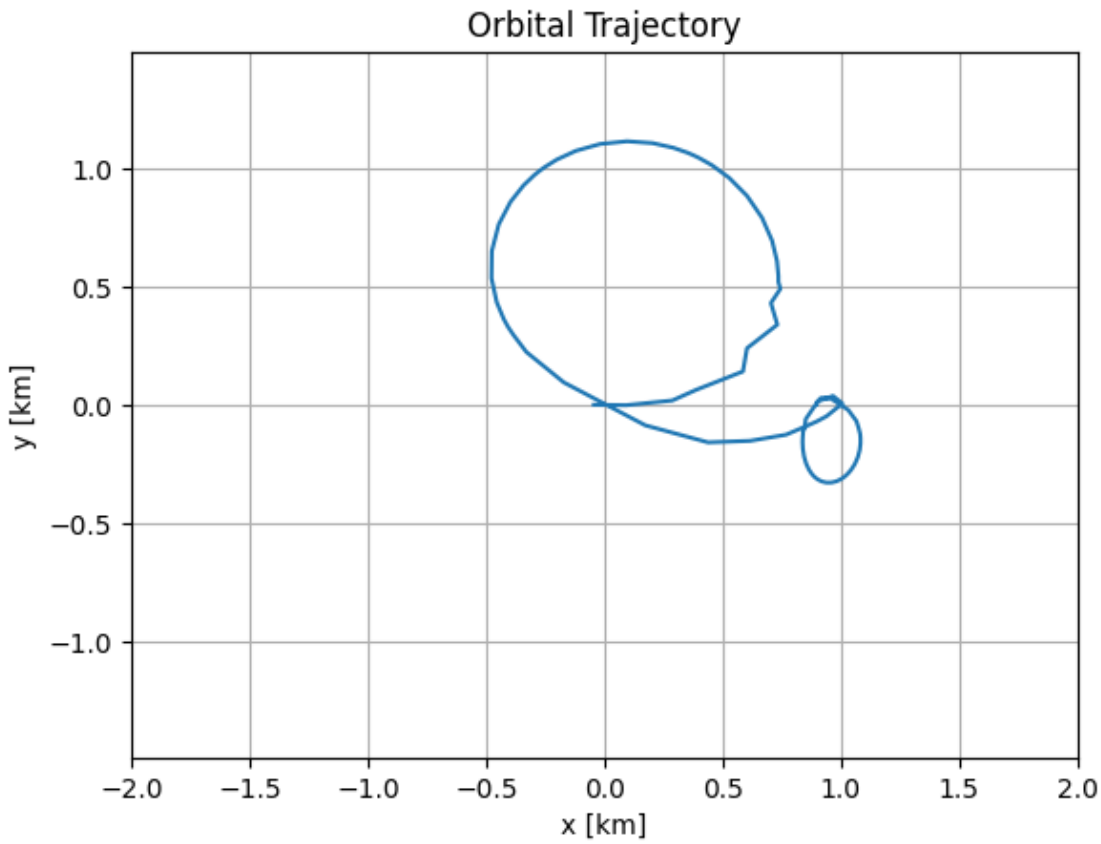

Figure 1: Attempt 1

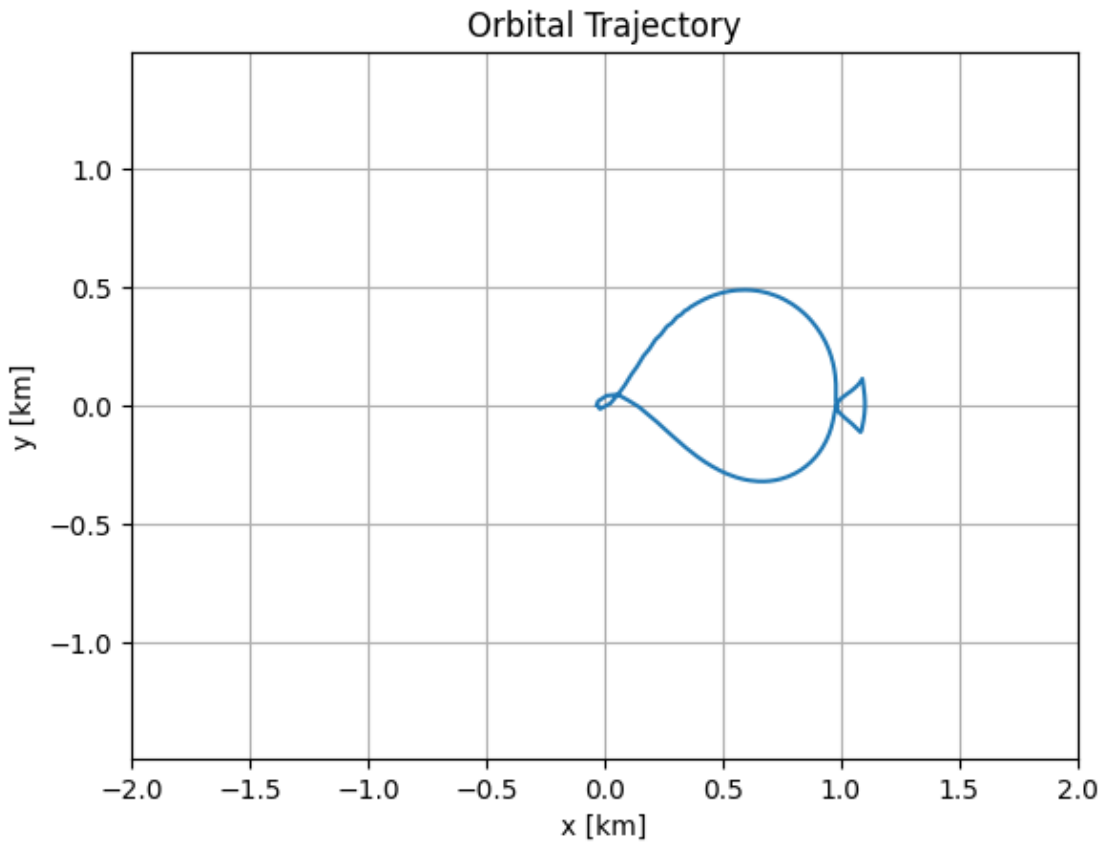

Figure 2: Attempt 2

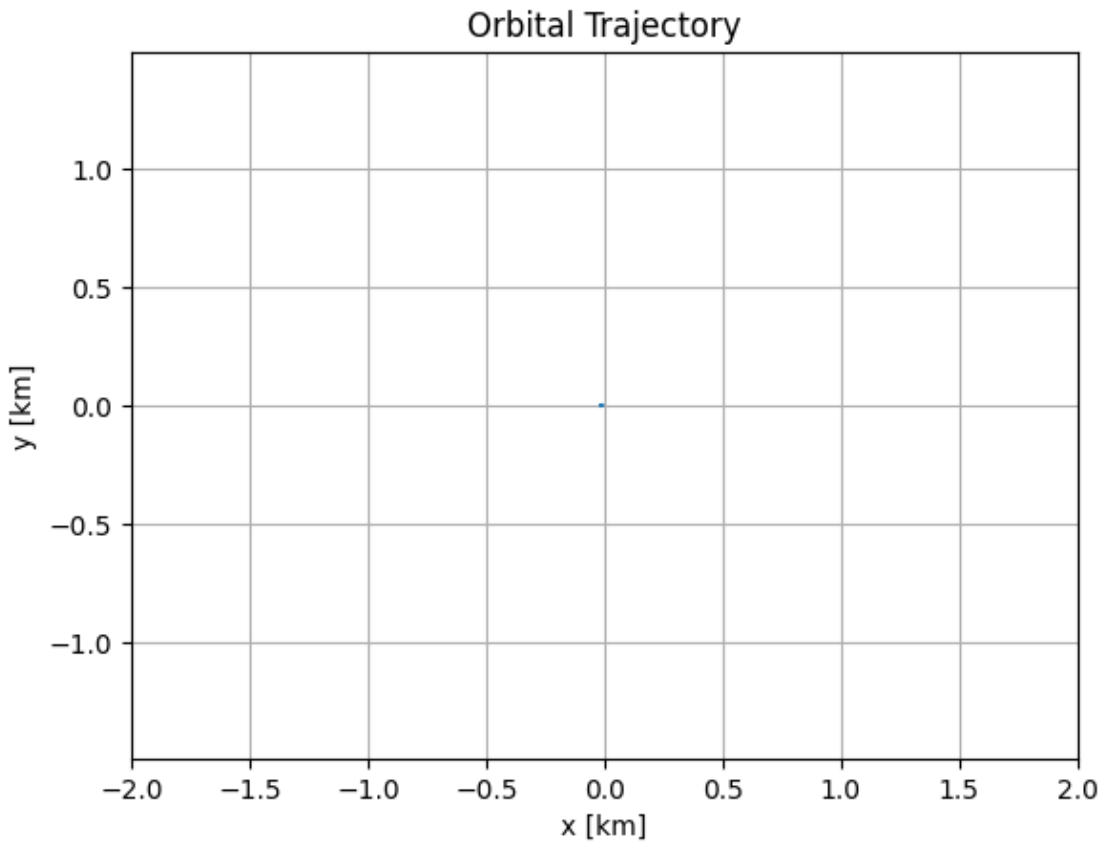

Figure 3: Attempt 3

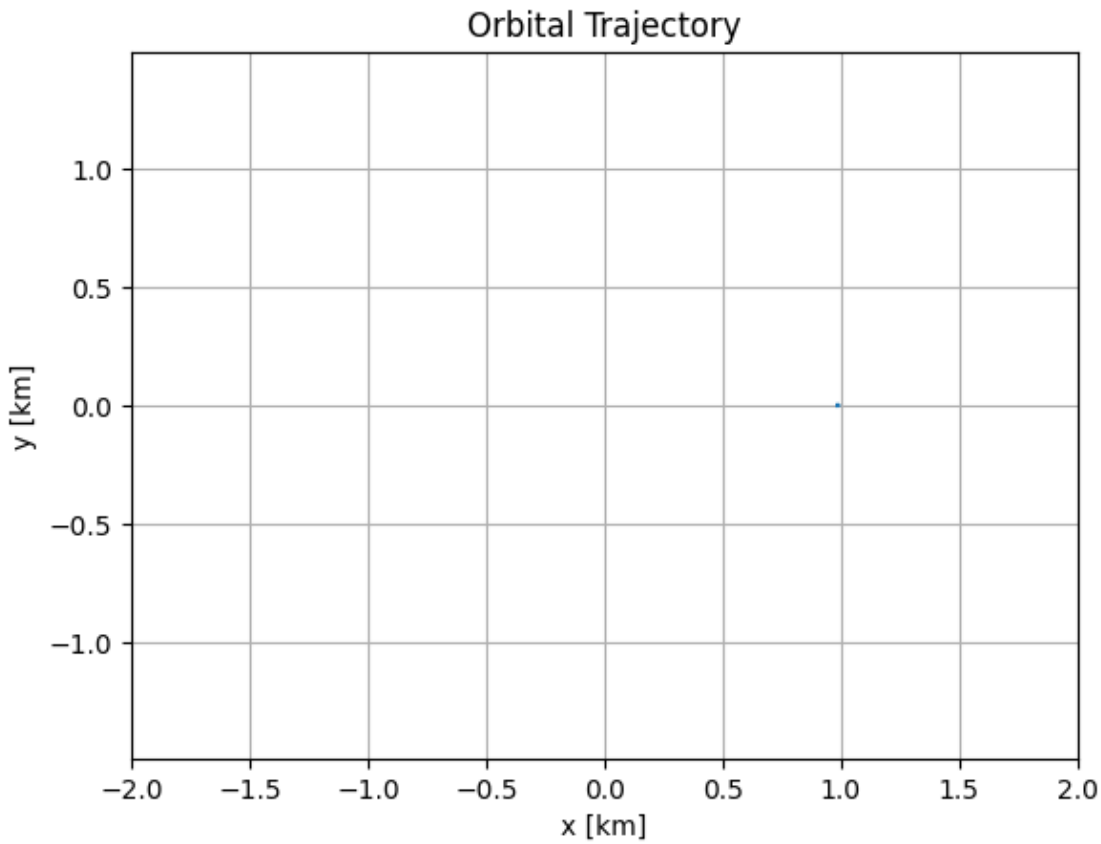

Figure 4: Attempt 4

3 Added Lower Time Bound

Realized that the final time was nowhere close, and with higher numbers of segments and points, only the trivial solution was found. Added lower time bound to fix.

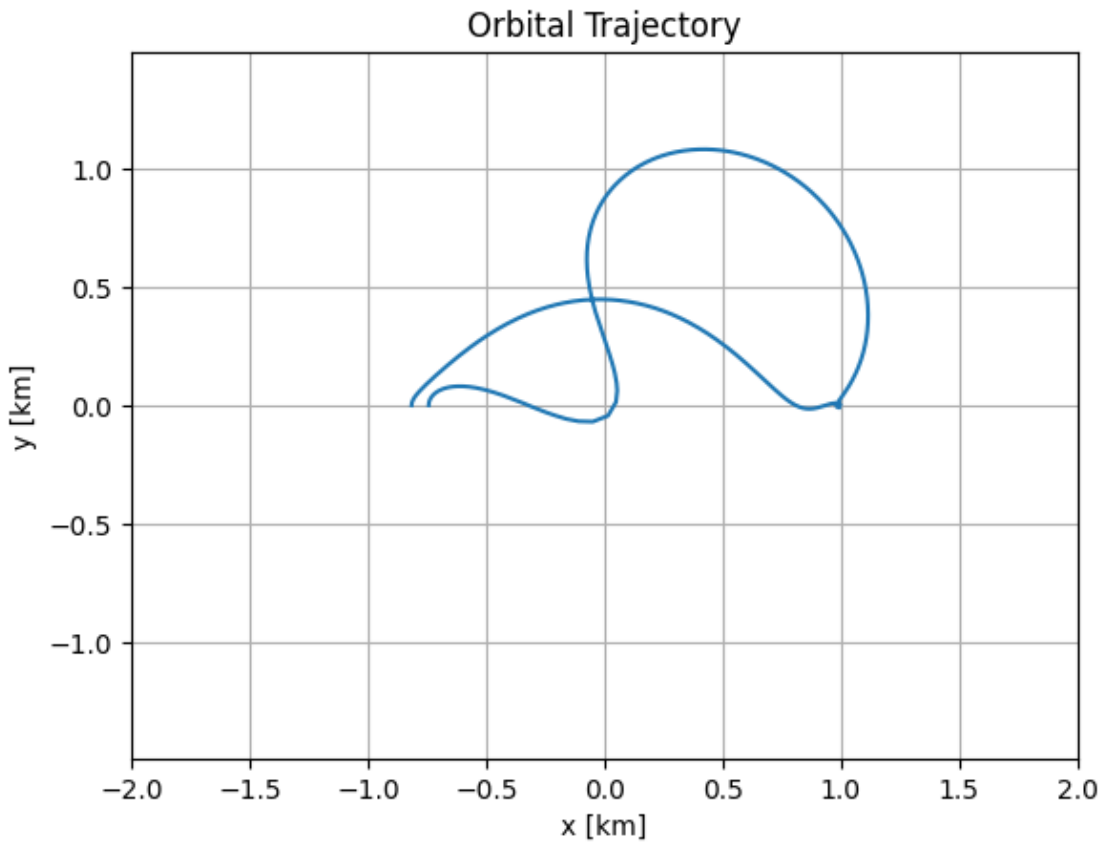

Figure 5: Attempt 5

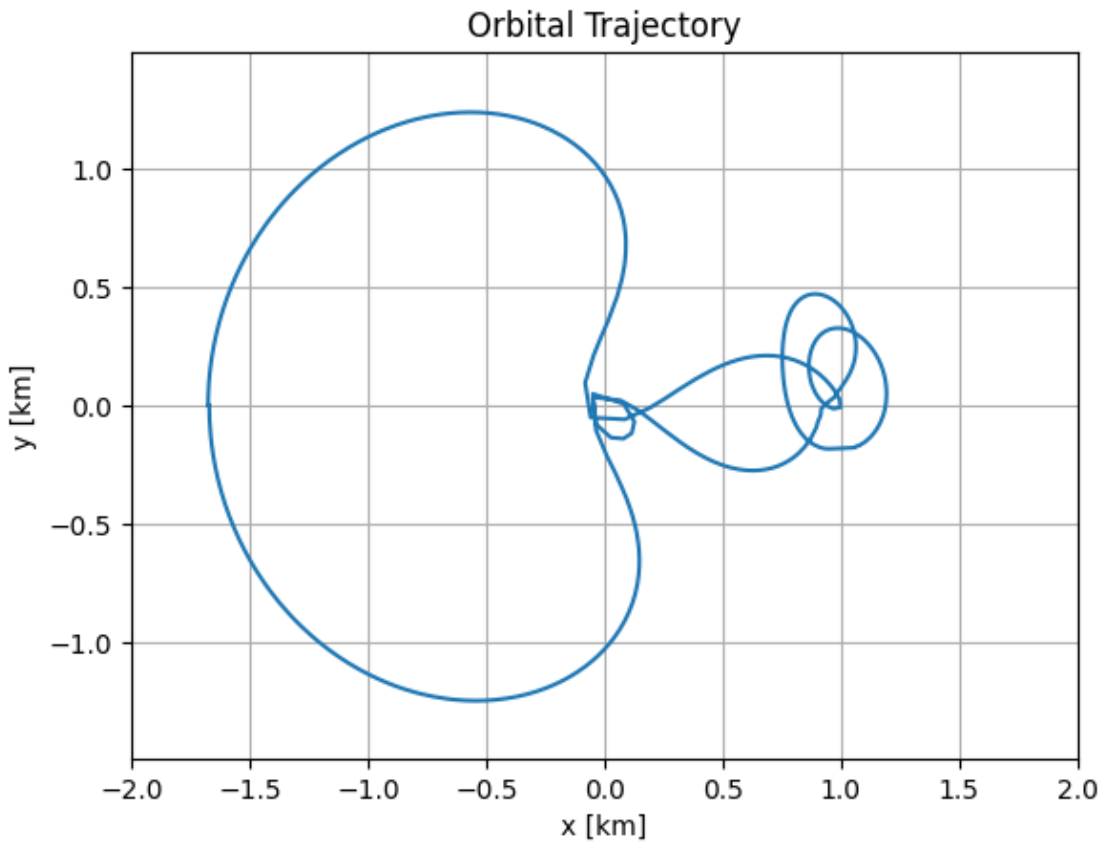

Figure 6: Attempt 6

4 Added Upper Time Bound

Added upper time bound and played with how tight the bounds were on the final time to the true period.

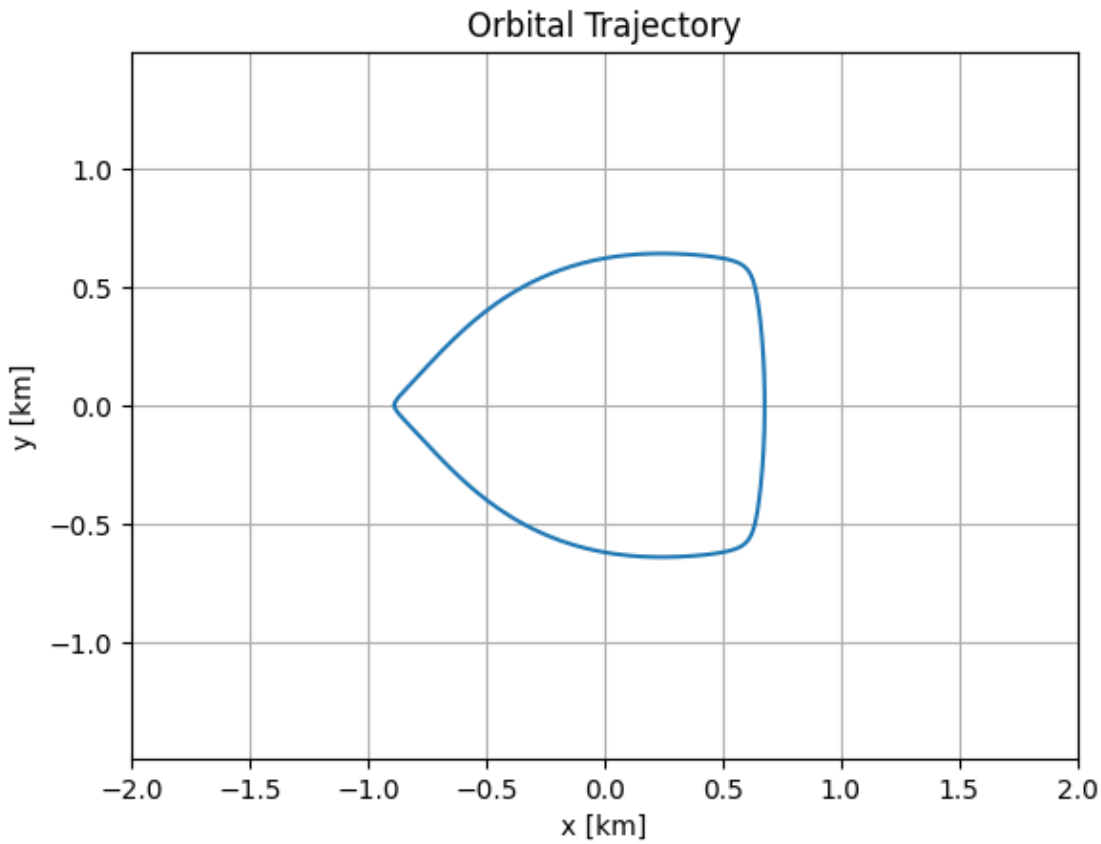

Figure 7: Attempt 7

Looks close, but period is >1 day off.

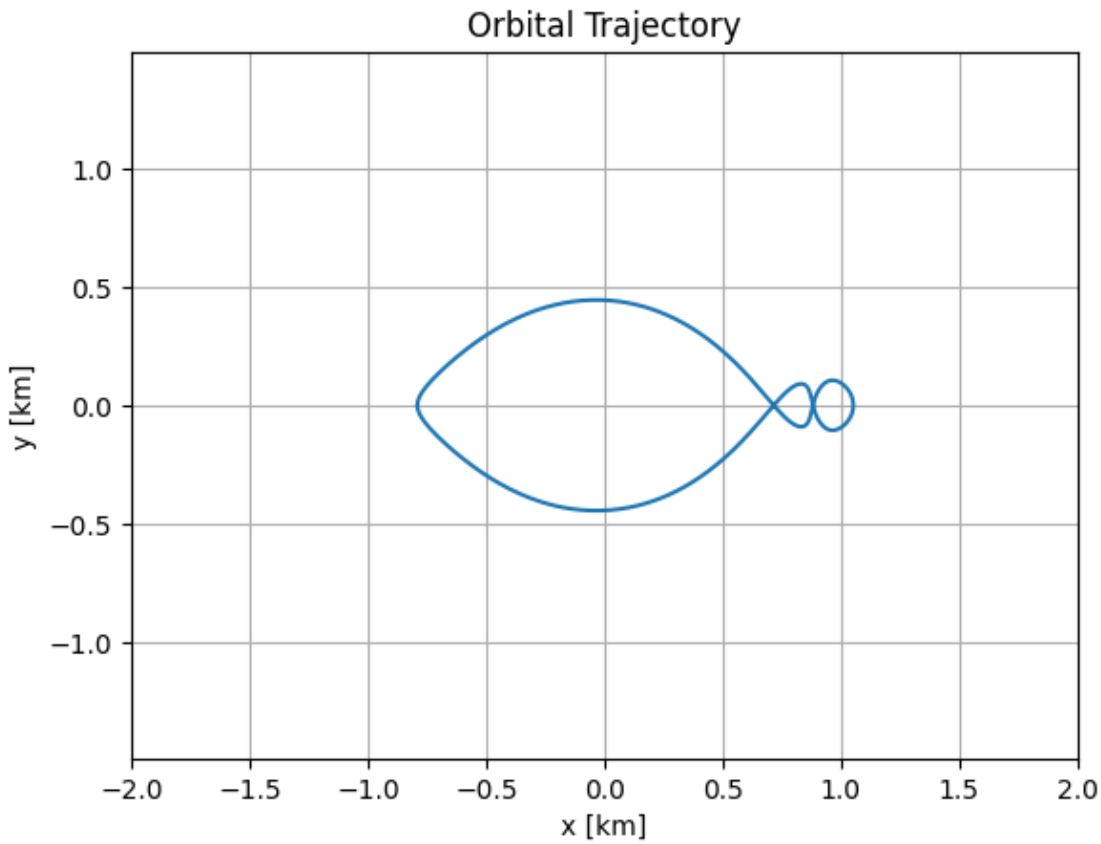

Figure 8: Attempt 8

5 Fine Tuning of Segments, Points, and Tightness of Bounds

Iterating on how bounds, segments, points, and tolerance.

If segments/points are too high, the bad velocity guesses ruin the solution.

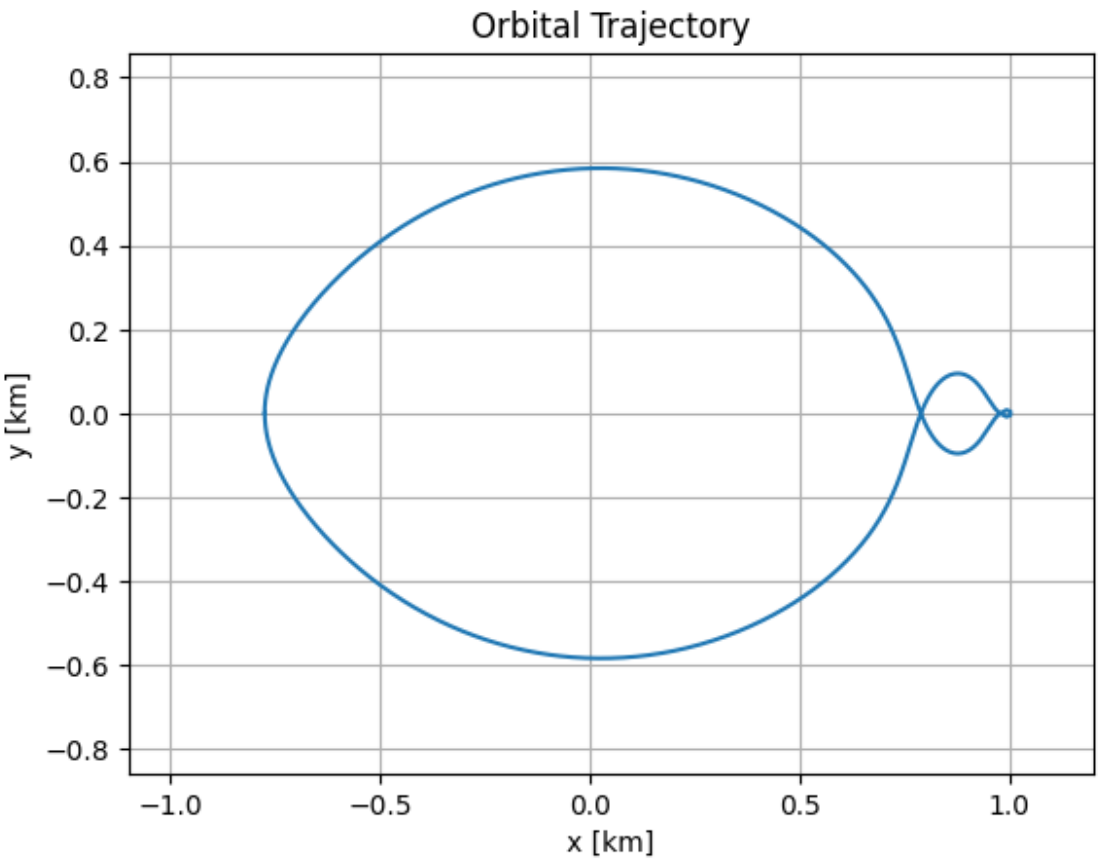

Figure 9: Attempt 9

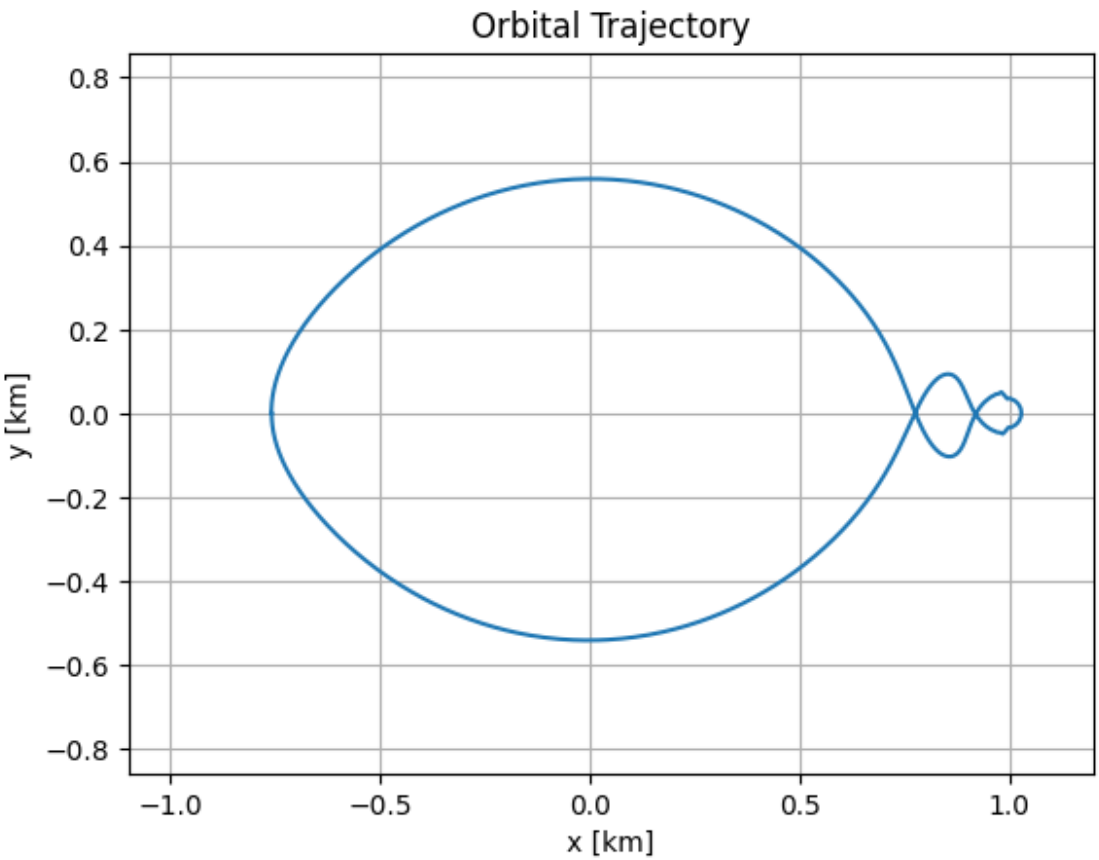

Figure 10: Attempt 10

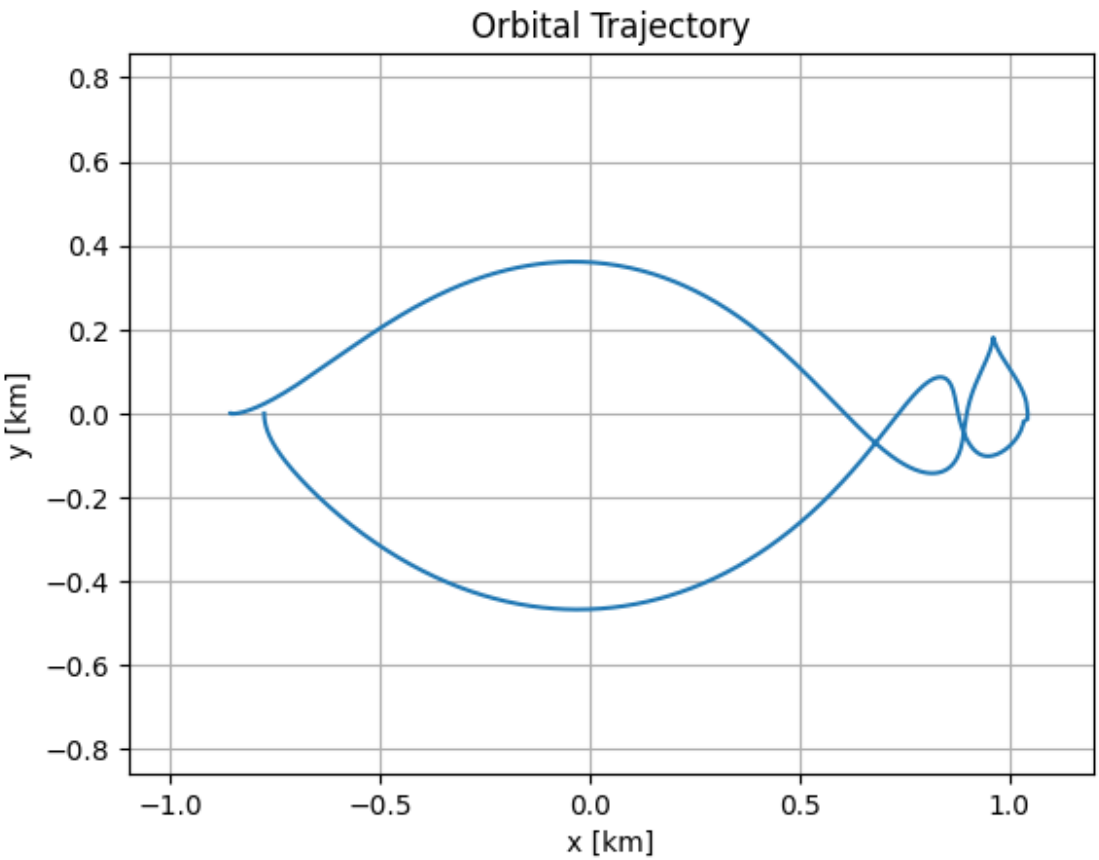

Figure 11: Attempt 11

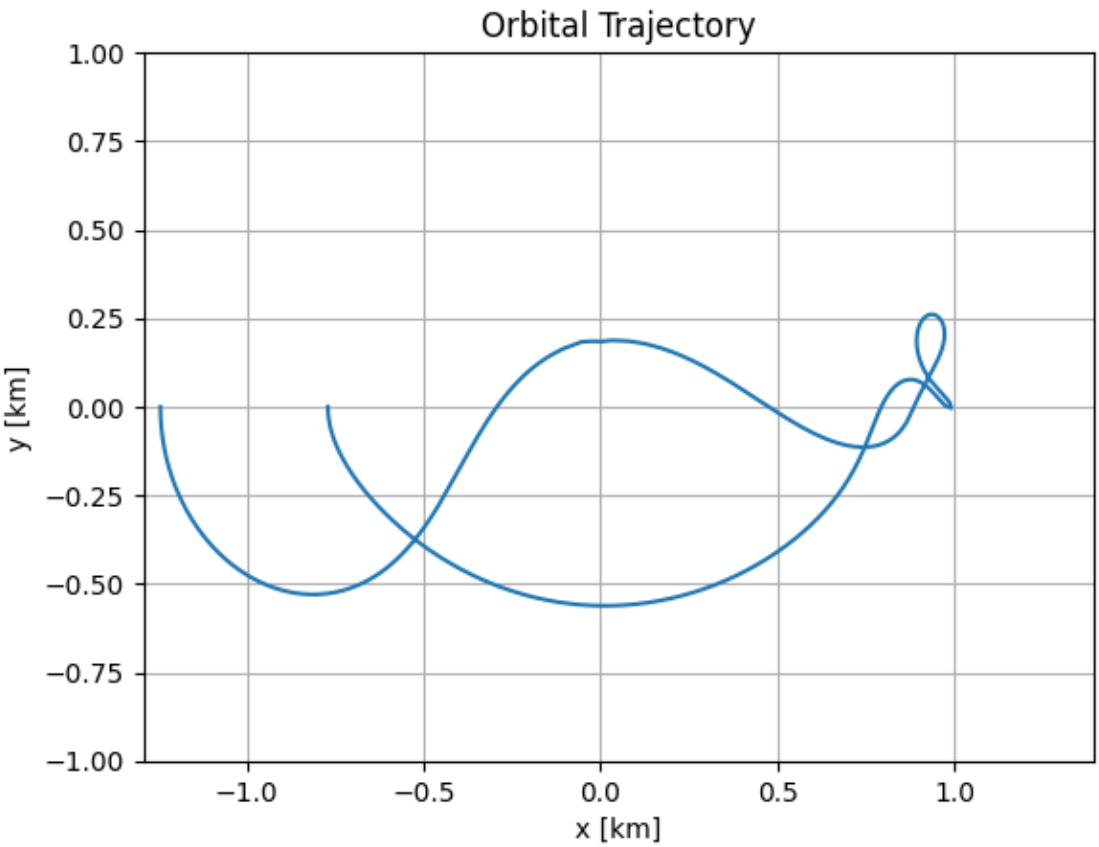

Figure 12: Attempt 12

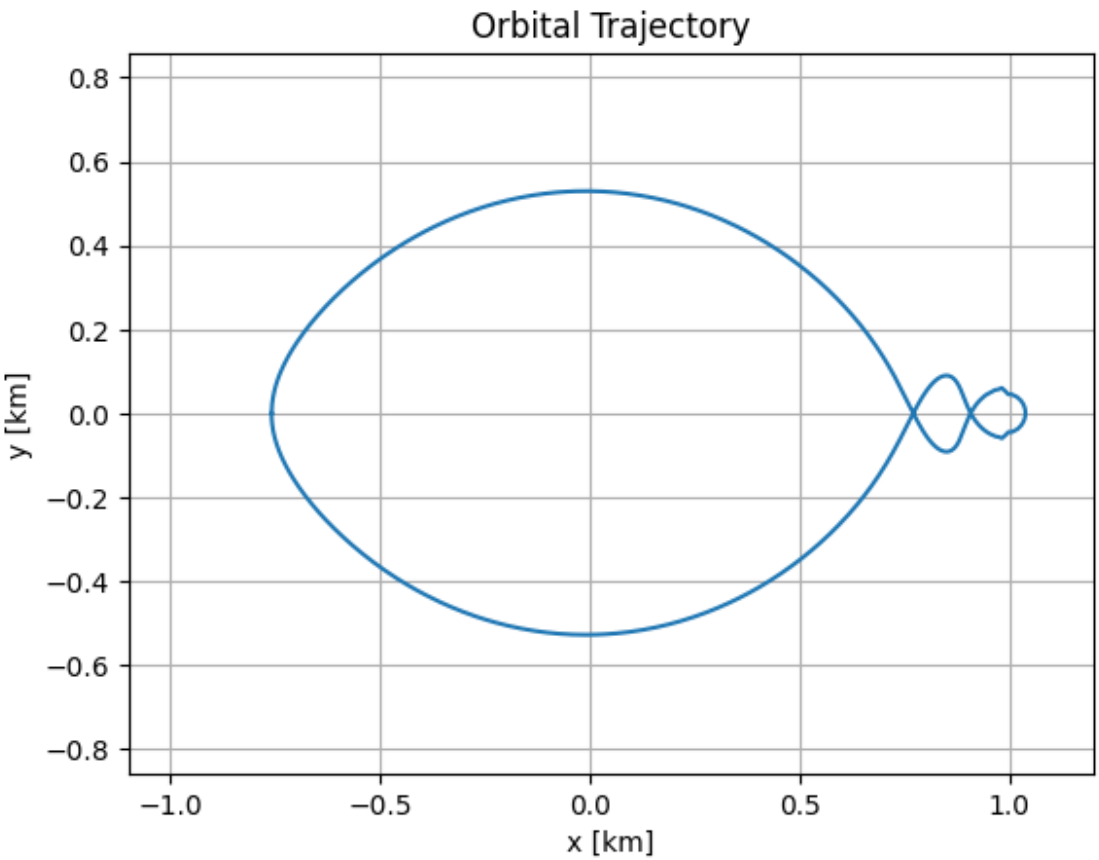

Figure 13: Attempt 13

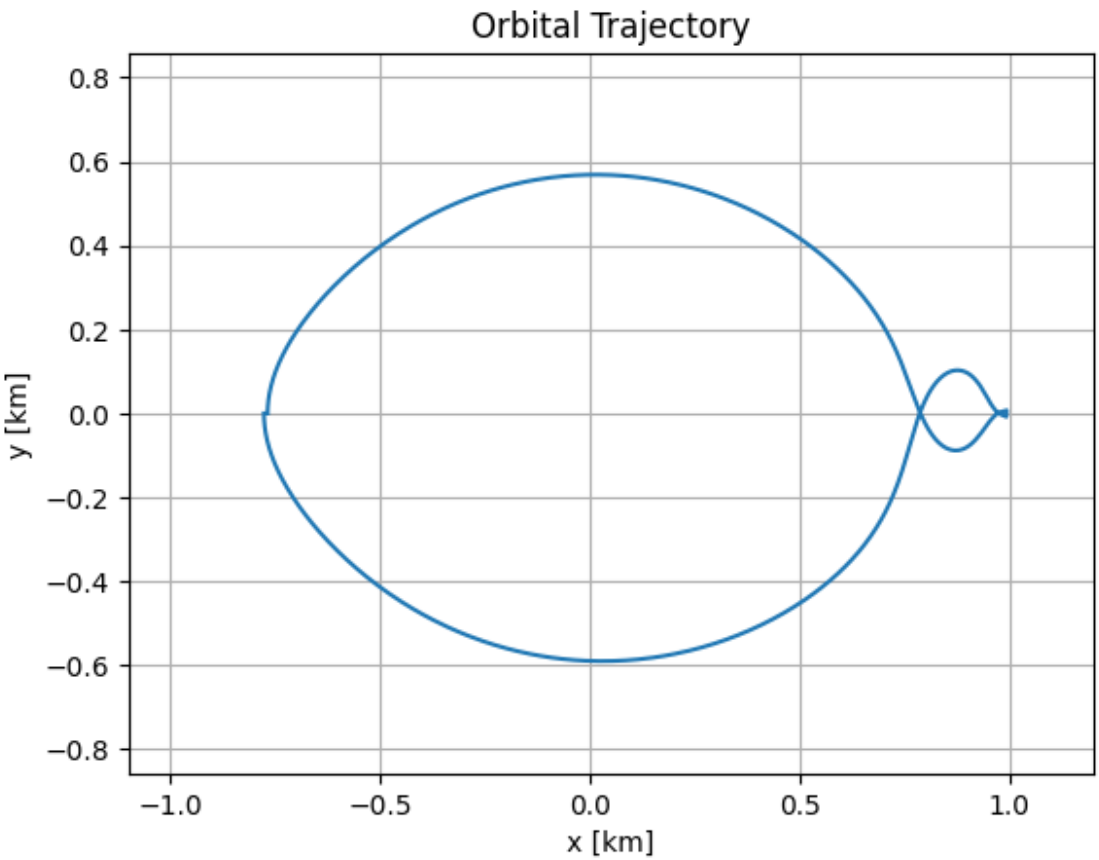

Figure 14: Attempt 14

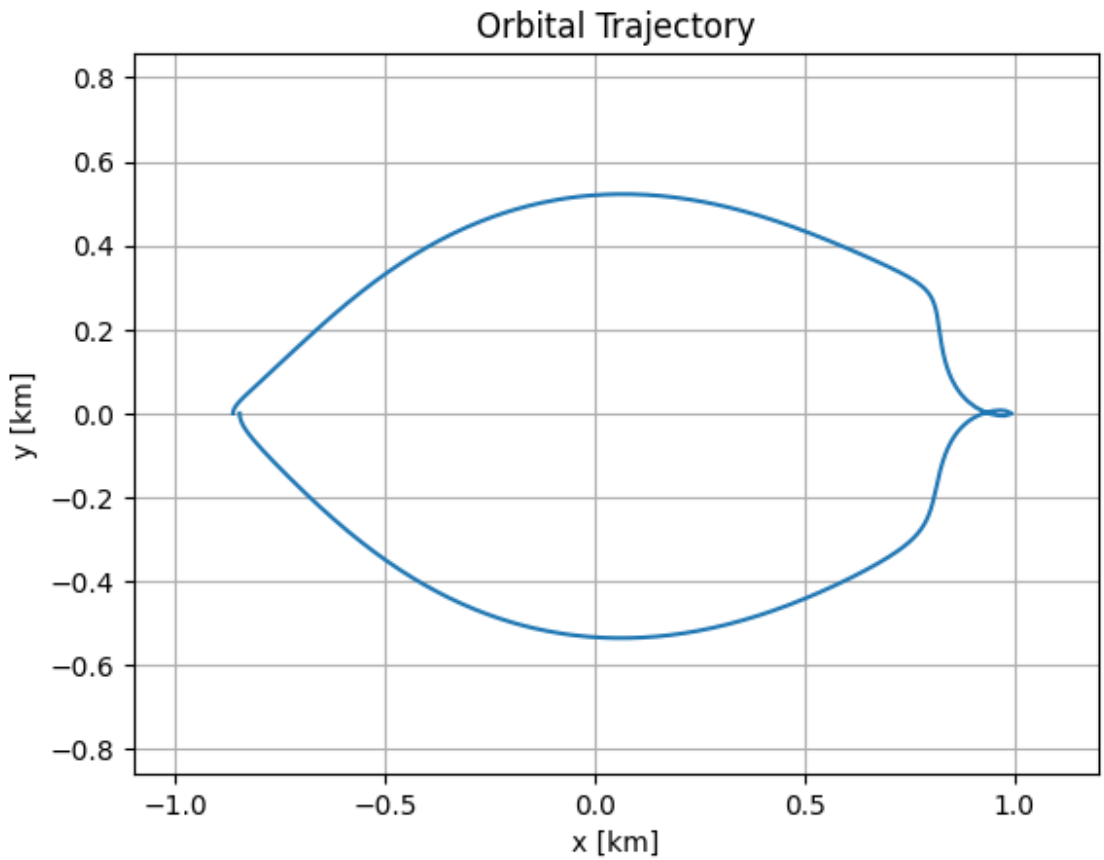

Figure 15: Attempt 15

Goldilocks zone of parameters to give a decent initial guess (5 hours off of actual period)

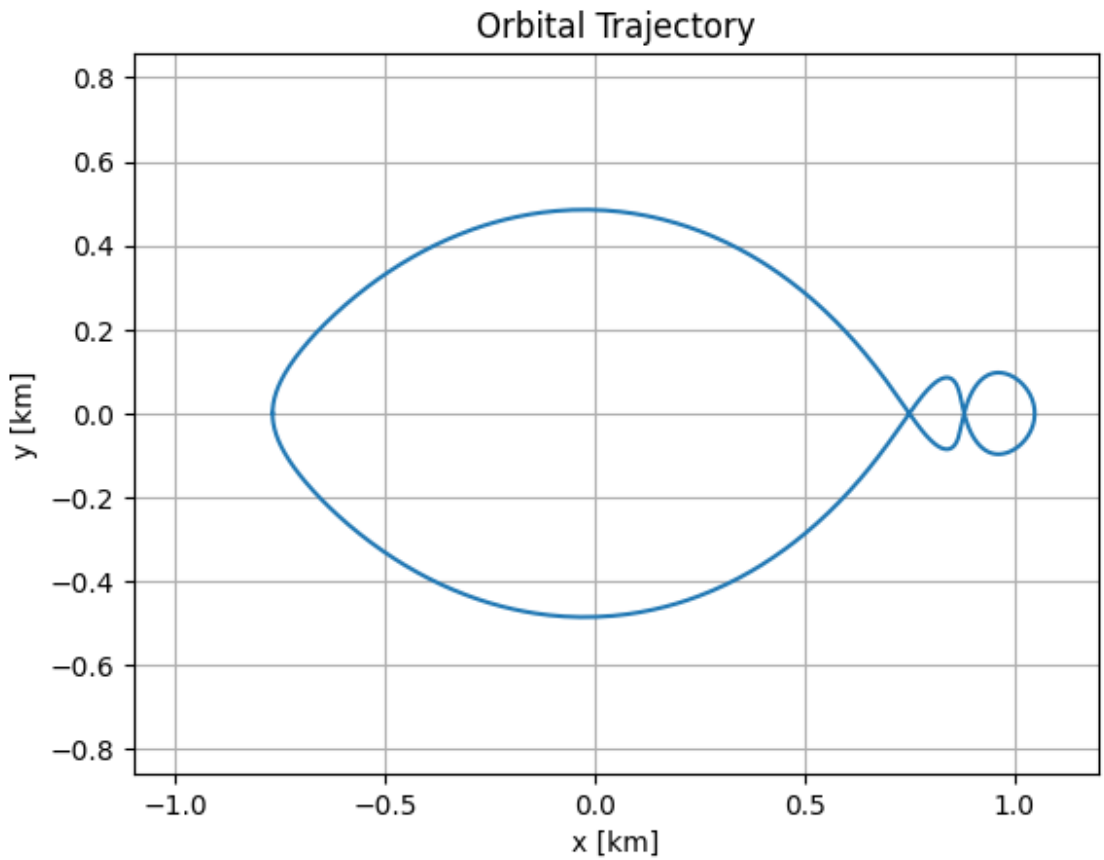

Figure 16: Attempt 16 (Taken as initial guess for $k_1 = k_2 = 1$ cyclers)

6 Further Refinement

Kept decreasing the tolerance.

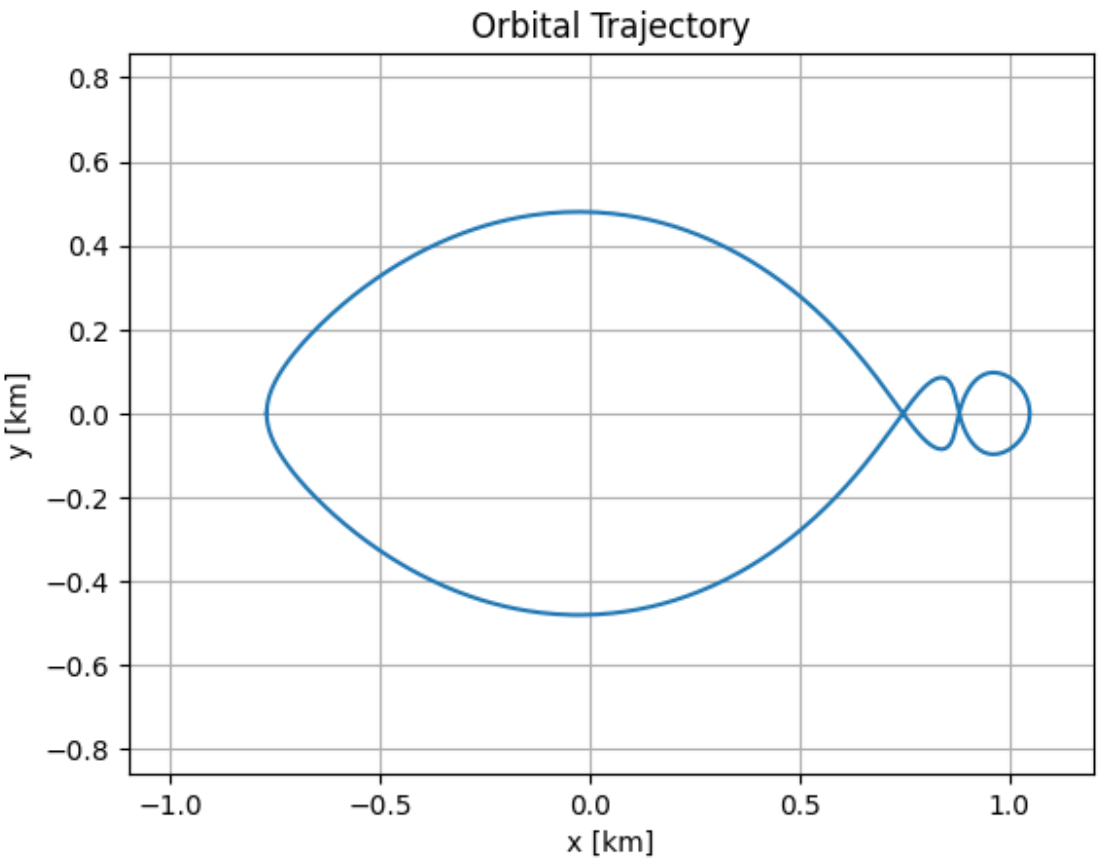

Figure 17: Attempt 17

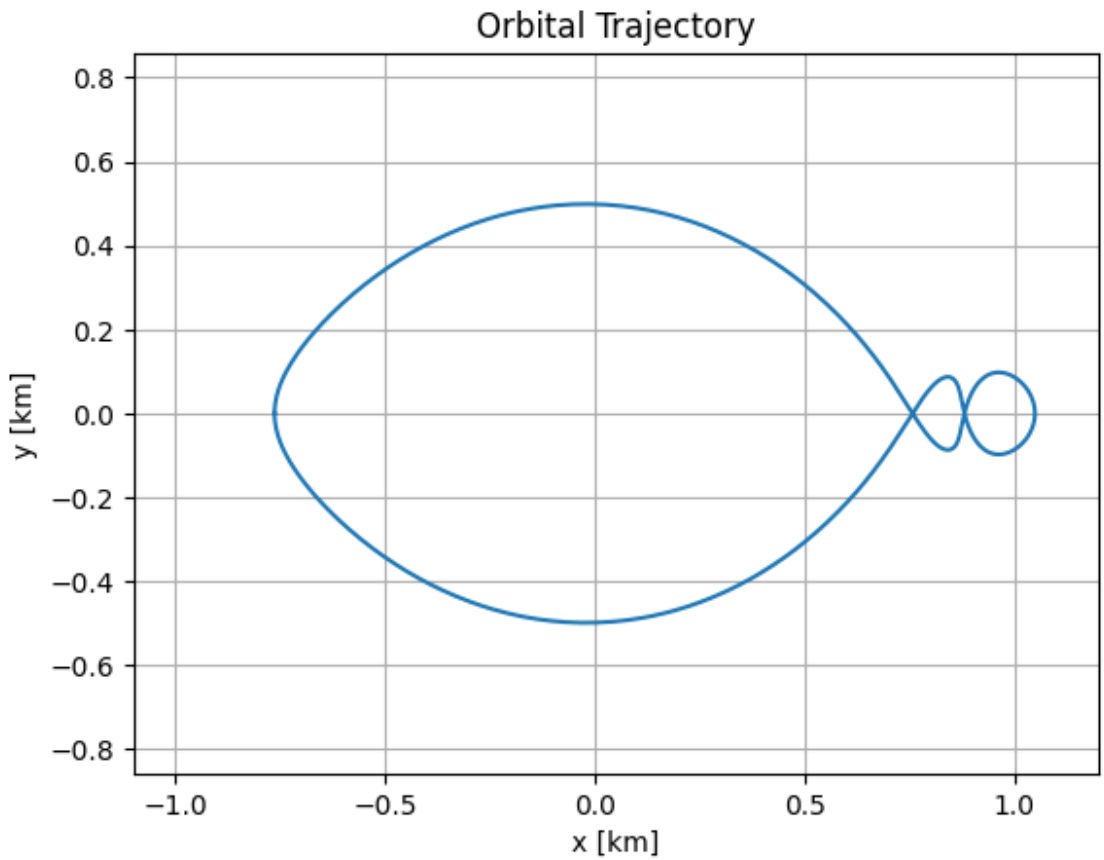

Figure 18: Attempt 18

Increasing segments and points from 30 to just 32 breaks the model.

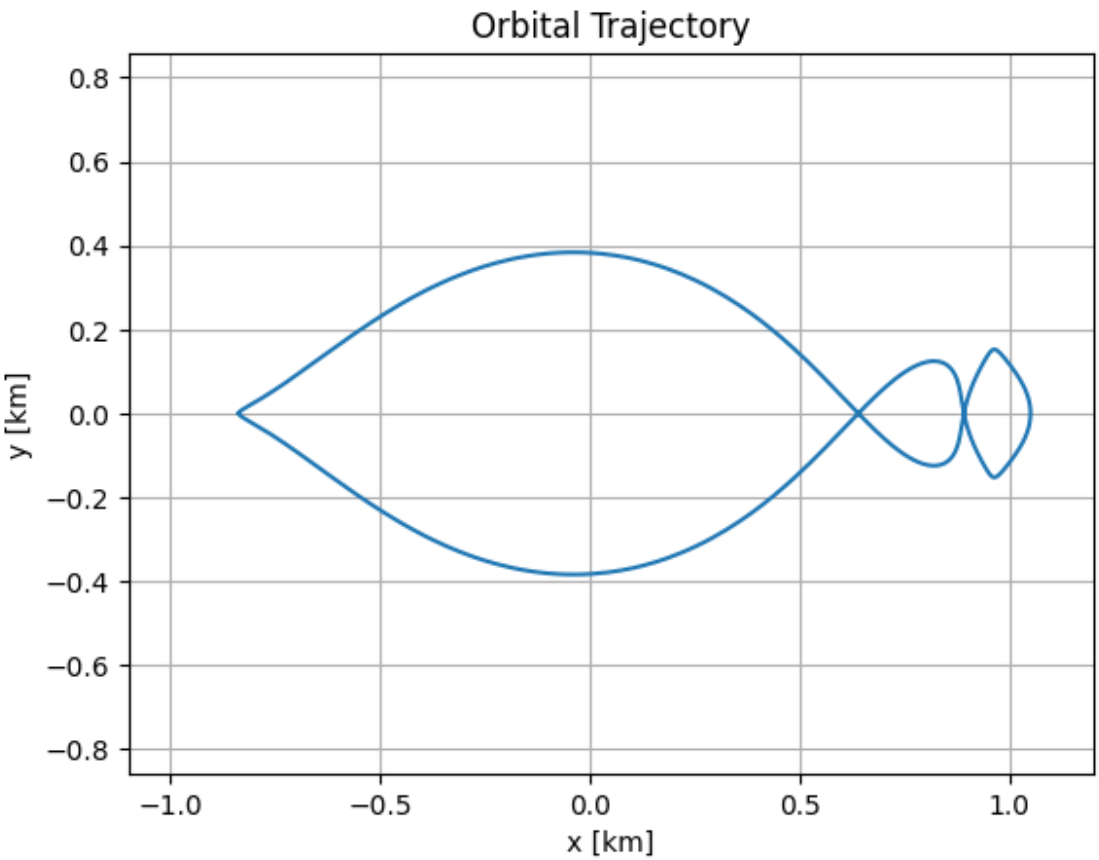

Figure 19: Attempt 19

7 Final Initial Guess

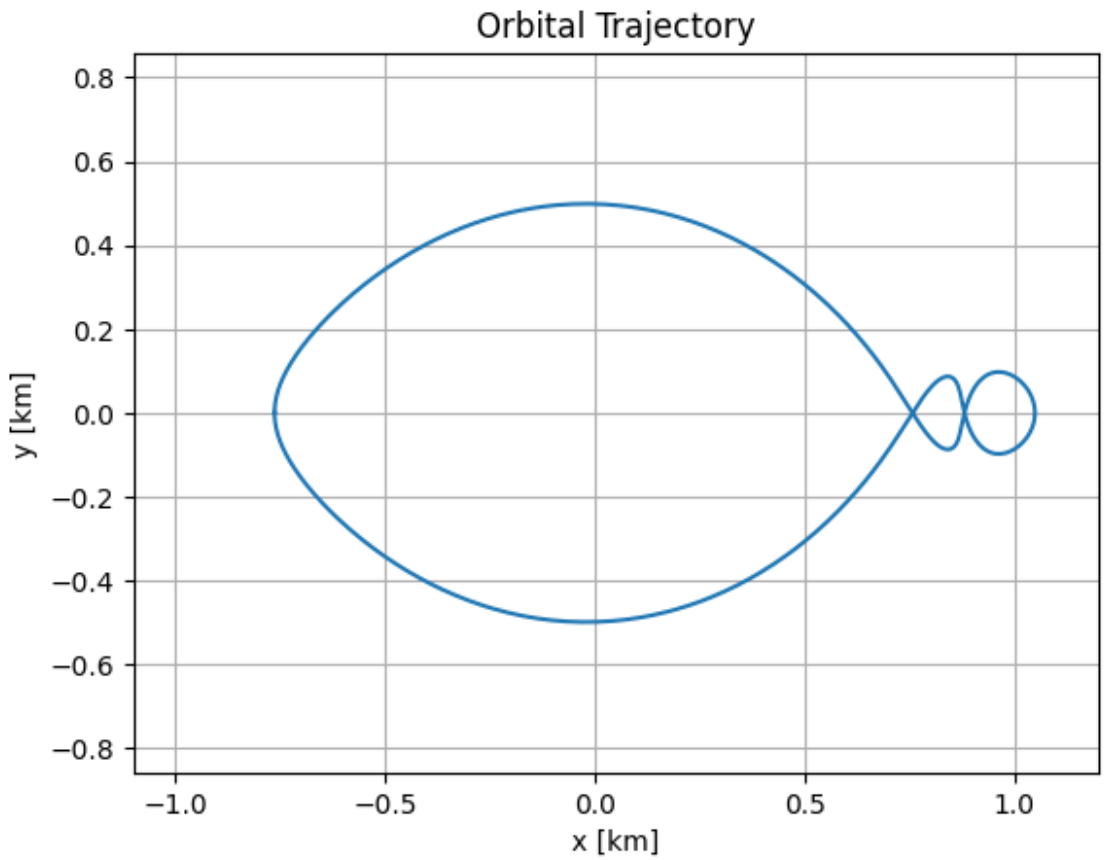

Figure 20: Attempt 20 (best initial guess)

7.1 Parameters

No objective function to minimize, set to just propogate the dynamics.

Orbital Parameter Guesses

- $x_i = -0.767856324800$
- $C = 3.151175879508174$
- $y = \dot{x} = 0$

Bounds

- $t_{i_{lower}} = t_{i_{upper}} = 0$
- $t_{f_{lower}} = 44.7538 - 1.5$ (further adjusted from days to time scale used in equations)
- $t_{f_{upper}} = 44.7538 + 1.5$
- discrete bounds = [0, 0, 0, 0]
- $\mathbf{x}_i[1:3] = \mathbf{x}_f[1:3] = [0, 0]$

Solver Parameters

- segments = points = 30

- ipopt tol = 1e-20

8 Final Initial Guess - Better

Removing the last point at the end of the time window as a guess improved the results. New generated orbit is 15 minutes off of stated period.

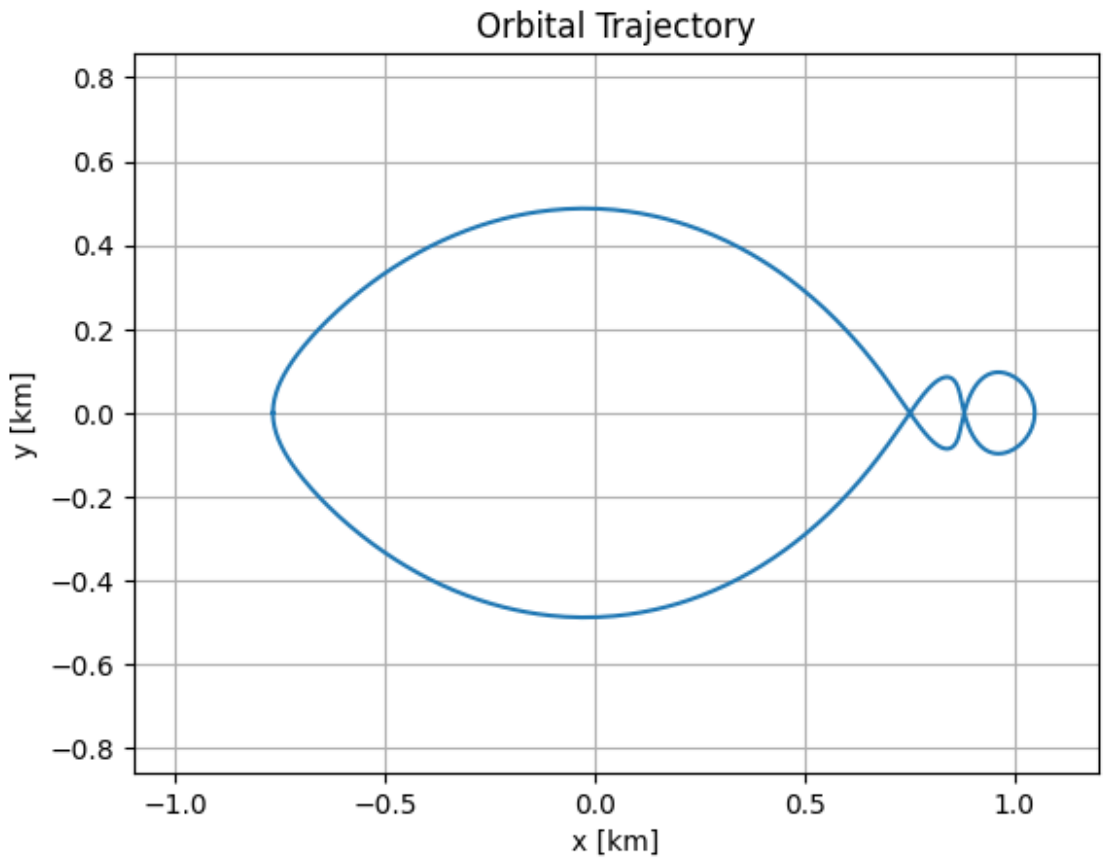

Figure 21: Attempt 21 (best initial guess again)

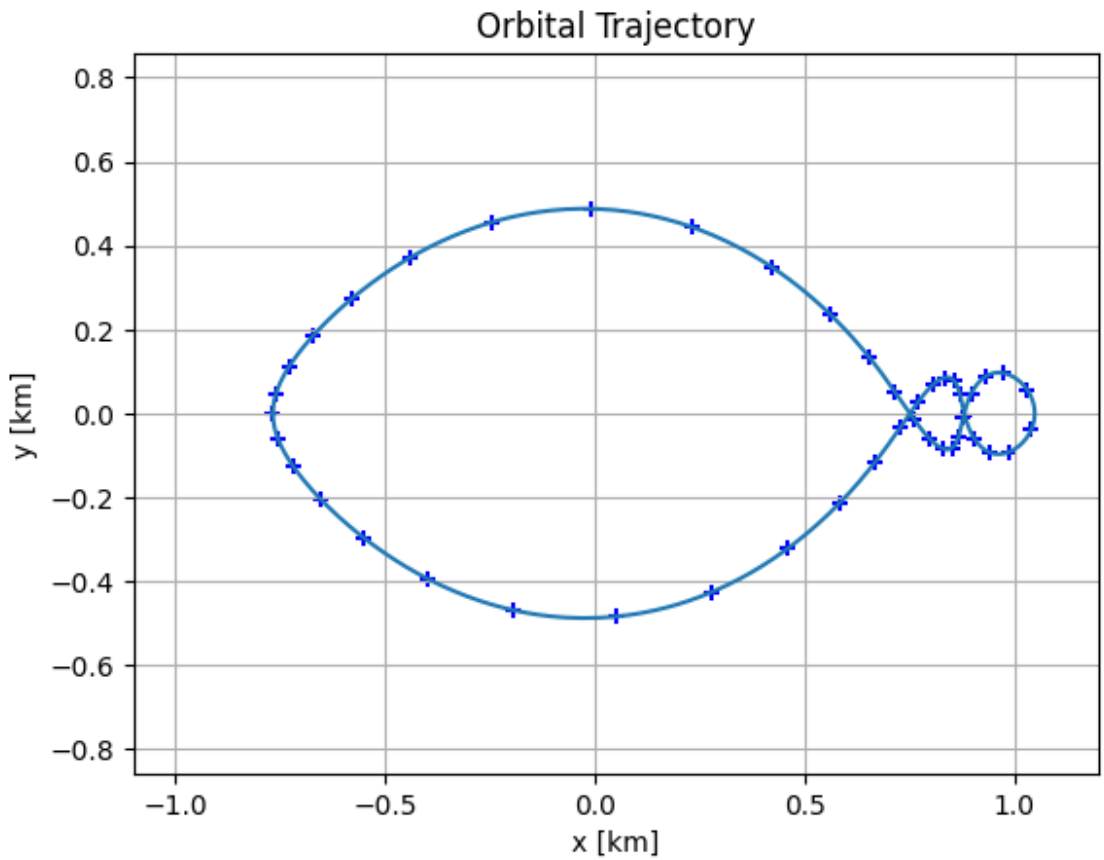

Figure 22: Attempt 21 with day markers

8.1 Parameters

No objective function to minimize, set to just propogate the dynamics.

Orbital Parameter Guesses

- $x_i = -0.767856324800$
- $C = 3.151175879508174$
- $y = \dot{x} = 0$

Bounds

- $t_{i_{lower}} = t_{i_{upper}} = 0$
- $t_{f_{lower}} = 44.7538 - 0.95$ (further adjusted from days to time scale used in equations)
- $t_{f_{upper}} = 44.7538 + 0.95$
- discrete bounds = [0, 0, 0, 0]
- $\mathbf{x}_i[1:3] = \mathbf{x}_f[1:3] = [0, 0]$

Solver Parameters

- segments = points = 30
- ipopt tol = 1e-100

Error from stated period: 0.024%